

# Tic Tac Toe Game in C

## Sample Output

### Program Startup

```
TIC TAC TOE  X O
```

- 1. Two Player
- 2. vs AI (Minimax)
- 0. Exit

Choice: 1

### 1. Two Player Mode

#### Initial Board

```
1 | 2 | 3
---|---|---
4 | 5 | 6
---|---|---
7 | 8 | 9
```

#### Match in Progress

Player X, enter position (1-9): 5

```
1 | 2 | 3
---|---|---
4 | X | 6
---|---|---
7 | 8 | 9
```

Player O, enter position (1-9): 1

```
0 | 2 | 3
```

|   |   |   |
|---|---|---|
|   |   |   |
| 4 | X | 6 |
|   |   |   |
| 7 | 8 | 9 |

Player X, enter position (1-9): 9

|   |   |   |
|---|---|---|
| 0 | 2 | 3 |
|   |   |   |
| 4 | X | 6 |
|   |   |   |
| 7 | 8 | X |

Player 0, enter position (1-9): 2

|   |   |   |
|---|---|---|
| 0 | 0 | 3 |
|   |   |   |
| 4 | X | 6 |
|   |   |   |
| 7 | 8 | X |

Player X, enter position (1-9): 1  
Position taken!

Player X, enter position (1-9): 3

|   |   |   |
|---|---|---|
| 0 | 0 | X |
|   |   |   |
| 4 | X | 6 |
|   |   |   |
| 7 | 8 | X |

## Player X Wins

Player 0, enter position (1-9): 4

|   |   |   |
|---|---|---|
| 0 | 0 | X |
|   |   |   |
| 0 | X | 6 |
|   |   |   |
| 7 | 8 | X |

Player X, enter position (1-9): 7

|       |  |   |  |   |
|-------|--|---|--|---|
| 0     |  | 0 |  | X |
| <hr/> |  |   |  |   |
| 0     |  | X |  | 6 |
| <hr/> |  |   |  |   |
| X     |  | 8 |  | X |

Player X WINS! 🎉

## 2. VS AI Mode

Choice: 2

You are X, AI is 0

### Human Move

|       |  |   |  |   |
|-------|--|---|--|---|
| 1     |  | 2 |  | 3 |
| <hr/> |  |   |  |   |
| 4     |  | 5 |  | 6 |
| <hr/> |  |   |  |   |
| 7     |  | 8 |  | 9 |

Player X, enter position (1-9): 5

### AI Move

AI thinking...

AI plays: position 1

|       |  |   |  |   |
|-------|--|---|--|---|
| 0     |  | 2 |  | 3 |
| <hr/> |  |   |  |   |
| 4     |  | X |  | 6 |
| <hr/> |  |   |  |   |
| 7     |  | 8 |  | 9 |

## Mid Game

Player X, enter position (1-9): 9

AI thinking...

AI plays: position 3

|       |  |   |  |   |
|-------|--|---|--|---|
| 0     |  | 2 |  | 0 |
| <hr/> |  |   |  |   |
| 4     |  | X |  | 6 |
| <hr/> |  |   |  |   |
| 7     |  | 8 |  | X |

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## Draw Match

|       |  |   |  |   |
|-------|--|---|--|---|
| 0     |  | X |  | 0 |
| <hr/> |  |   |  |   |
| 0     |  | X |  | X |
| <hr/> |  |   |  |   |
| X     |  | 0 |  | X |

DRAW!

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## Invalid Move Handling

Player X, enter position (1-9): 5

Position taken!

Player X, enter position (1-9): 8

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## Exit Program

1. Two Player
2. vs AI (Minimax)
0. Exit

Choice: 0

Program terminated successfully.

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## Features Demonstrated

- Two Player Mode
- AI Opponent Using Minimax Algorithm
- Smart Move Selection
- Win Detection (Rows, Columns, Diagonals)
- Draw Detection
- Invalid Move Handling
- Interactive Console Interface
- Turn-Based Gameplay

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## Board Position Reference

|       |  |   |  |   |
|-------|--|---|--|---|
| 1     |  | 2 |  | 3 |
| <hr/> |  |   |  |   |
| 4     |  | 5 |  | 6 |
| <hr/> |  |   |  |   |
| 7     |  | 8 |  | 9 |

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